

A Finger-Tracking System for Studying Braille Reading on a Tablet

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Abstract—In order to investigate braille reading behavior, we need a device capable of tracking readers' finger movements in real-time. This paper presents software for tracking the fingers of braille readers. The software is developed to observe reading braille-embossed paper on an Android tablet. It provides online visual information of finger movement patterns, reading durations on cells, total reading time, and the average reading rate. This allows researchers to investigate braille reading in more depth in order to help improve the readers' braille reading skills.

Keywords—assistive computing; braille reading; educational technology; mobile computing

I. INTRODUCTION

Currently there are many devices to assist the blind to read, such as JAWs, the most popular screen reader using text-to-speech conversion for people with visual impairments. A recent iPhone application using the augmented reality (AR) technology, Aipoly, helps the visually impaired to “see” including to read text and recognise objects. Nevertheless, literacy is one of the most important skills to facilitate cognitive development and maintain cognitive process into old age [1]. Moreover, Ryles' [2] found that people with visual impairment tended to have higher levels of employment if they were able to read braille.

Few studies have investigated braille reading behavior in comparison to sighted reading, since braille readers are a minority and not many technologies support braille reading research. The first device that explored hand movements of braille readers was built by Bürklen [3] in 1932 and called a “Tastschreiber.” The readers' finger was connected to a pen-like instrument that scratched a trace of the finger movement on smoked paper. Most studies before 2010 used a video camera to record braille readers' hand movements in which the sampling rates were between 25 to 40 frames per second [4]. Most of these studies involved the manual analysis the finger position data from each video frame. This technique took time and the sampling rate was quite low. In 2011, Hughes [5] built an apparatus for tracking the movement kinetics of braille readers using a digitising tablet, in which the sampling rate of finger position was 100 Hz. Aranyanak and Reilly [6] devised a high-resolution finger-tracking system using a Wiimote to examine finger movements in braille readers. Both these devices are inconvenient for researchers to set up. This study presents an application for

tracking finger movements in braille reading that is easier to use to explore the reading behavior of braille readers on embossed-braille paper with an Android tablet.

Recently, a report showed the number of people who live with vision impairment across the world was around 253 million. Of these, 36 million are totally blind and the rest have moderate to severe visual impairments [7]. In Thailand, the report on the situation of disabled in Thailand showed that there were 188,050 Thai people with sight loss, which represents a third of all disabled. There are 12 schools for the blind in Thailand; the biggest center is located in Bangkok. Less than 10% of Thai blind have a chance to get education and job training. Less than 5% are employed in a stable position. Most Thai blind earn money by selling lottery tickets, which is supported by the government.

In the past, research on braille in Thailand focused on developing software to convert Thai into braille and vice versa (e.g., Foundation for the blind in Thailand, 2008 [8]). Recently, a group of researchers for the blind in Thailand produced software to produce audio books, such as, “Read for the blind” by Pongsawat [9]. Samsung in Thailand created a special ink called “Touchable Ink” that can be used as an alternative to a braille printer. The ink can be used with home printers and the price of printing braille per page is much cheaper than using a standard braille printer. However, there are few studies focused on braille reading behavior and none of them used an online system to record readers' fingers during braille reading.

The research described here involves developing software to track fingers on embossed braille using an Android tablet. This system records finger movement online and provides a report of reading duration on cells, total reading time, and the average reading speed. The report can be exported to an excel file so that further analysis can be done.

II. THE BRAILLE WRITING SYSTEM IN THAI

Braille is a system that enables people with visual impairments to read and write through touch. A braille cell on paper comprises six dots allowing for 64 distinct patterns. The English braille system includes contractions, helping to reduce the size of braille books. Thai script is a syllabic alphabet. There are 44 consonants, 32 vowels, and 4 tone markers. Thai braille system does not use contractions, and also the writing system itself contains tone markers and some vowels located above or below a character. In the braille

version they have to be located on a single line in sequence. For example, “รู้ณ” is written in braille as follows: ၃/၁/၇/၇/၇; the braille codes are 1235/ 256/ 16/ 1345. Consequently, Thai braille is read more slowly than English braille, which has a reading rate of 100 words per minute [10]. Srisawat [11] observed the patterns of hand activity and reading speed of Thai braille. The study found that over 73% of Thai braille readers preferred to use both hands more than only one hand in reading. The average reading rate of two-handed readers was 41 WPM and 35 WPM for one-handed readers. In 2015, Jitsodsakul, Niyomphol, and Sirirungruang [12] compared the reading speed between sighted readers and readers with visual impairments and found that the average reading rate of sighted readers was twice as fast than those who read braille: 122 WPM in print and 52 WPM in braille. However, none of these studies used an online system to record the reading time, so their reading rates may not be entirely accurate.

III. FINGER TRACKING SYSTEM

The application was developed using Java on Android Studio. The finger tracking system is comprised of the following devices.

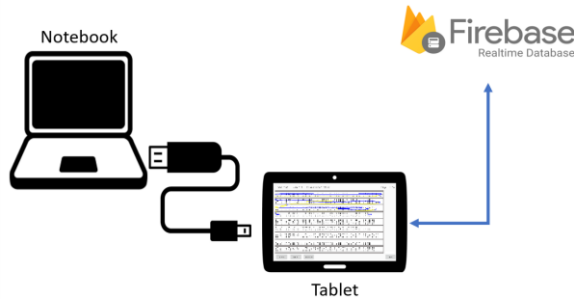


Figure 1. Finger tracking system.

Figure 1 presents the finger tracking system. A notebook, running on ASUS K46CB, Intel Core i7-3537U (3.10 GHz), is used to create the software. The coding was done in Android Studio using Java. Braille characters showed as pictures on the tablet screen are imported into the drawable library on the notebook. Then the application can be transferred into a tablet, Lenovo IdeaTab S6000 (CPU: MTK 8389 QC 1.2GHZ), using a USB cable. The tablet comes with a 10.1” display with a resolution of 1280 x 800 pixels. Once the program is uploaded to the tablet, a researcher can bring just the tablet to collect data. Transparent braille paper is placed on the tablet screen using sticky tape on four edges. The data recorded on the tablet is transferred to a database on this link: <https://console.firebase.google.com>. Firebase is used to process the data in real time. A text file can be downloaded from the cloud database.

Figure 2 shows the overall algorithm of the tracking system from the beginning to the end. A key feature of the algorithm is the facility to change the color of the finger trace to allow the visualization of re-reading. Thus, the instructor can immediately see the areas of the text where the reader is having difficulty.

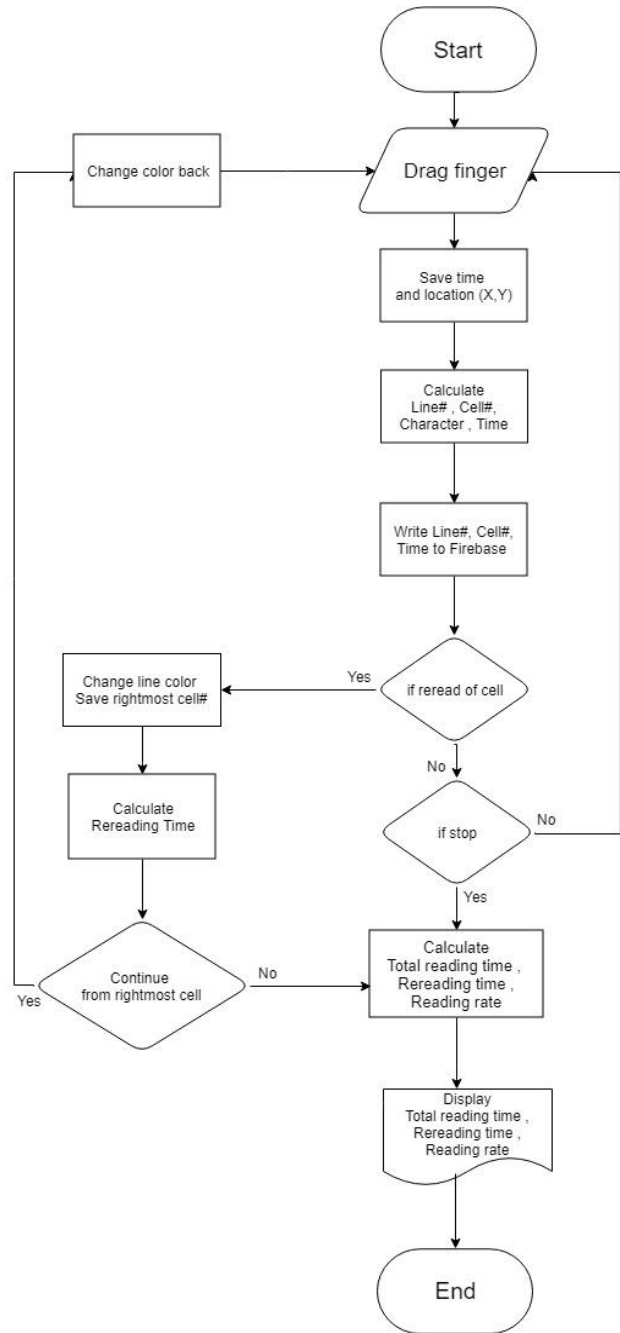


Figure 2. Flowchart of the algorithm of the tracking system.

IV. APPLICATION

This software can detect finger movements on a tablet screen. However, this application is a prototype, and can now just track two fingers. The reason we use only two fingers at the moment is because most braille readers only use two fingers, usually the left and right index fingers as the main fingers in reading. However, the program can track up to five fingers and the code can be modified easily to accommodate this. A maximum of 34 cells per line and a maximum of 9 lines per screen can be displayed on a 10.1”

display with a resolution of 1280 x 800 pixels. However, these can be changed depending on the size of the tablet screen used. The software supports reading multiple pages. A sample screen of the application is showed in Figure 3.

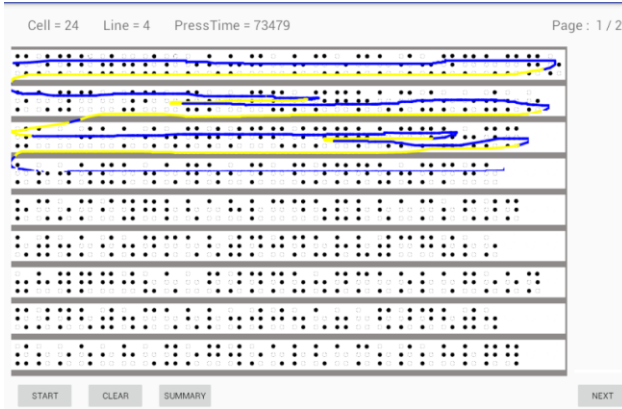


Figure 3. One-finger tracking.

Figure 3. shows one-finger tracking during braille reading. The line is recorded when a finger presses on the screen. The blue line represents reading from left to right. The yellow line represents return sweeps from the end of one line to the beginning of the next. Return sweeps will be deleted when computing the reading time. The application can show cell, line, and page numbers on the screen. “PressTime” is the current reading duration. The “start” button is for starting a recording of the finger movements. The “clear” button is for clearing the line on the screen. The “summary” button is for showing a summary page of reading statistics. This includes total reading time in milliseconds, total reading time excluding regressions, total rereading time, and reading rate per character. The “next” button is for moving to a new page.

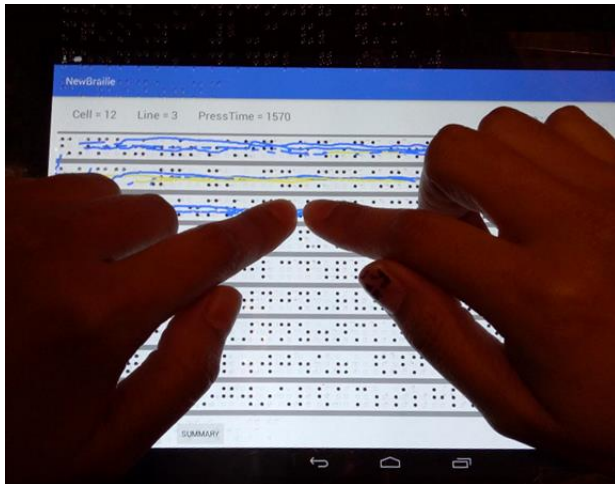


Figure 4. Two-finger tracking.

Figure 4. illustrates two-finger tracking. The following code is for tracing fingers while reading a transparent braille paper on the tablet screen.

```
public boolean onTouchEvent(MotionEvent event)
{ //get X and Y
  float x = event.getX();
  float y = event.getY();
  //Create and set count of finger : Start
  int pointerCount = event.getPointerCount();
  int cappedPointerCount = pointerCount > MAX_FINGERS ?
    MAX_FINGERS : pointerCount;
  int actionIndex = event.getActionIndex();
  int action = event.getActionMasked();
  int id = event.getPointerId(actionIndex);
  if ((action == MotionEvent.ACTION_DOWN || action ==
    MotionEvent.ACTION_POINTER_DOWN) && id < MAX_FINGERS)
  { mFingerPaths[id] = new Path();
    colorsMap.put(mFingerPaths[id], Color.BLUE);
    mFingerPaths[id].moveTo(event.getX(actionIndex),
      event.getY(actionIndex));
  }
  else if ((action == MotionEvent.ACTION_POINTER_UP || action ==
    MotionEvent.ACTION_UP) && id < MAX_FINGERS)
  { mFingerPaths[id].setLastPoint(event.getX(actionIndex),
    event.getY(actionIndex));
    mCompletedPaths.add(mFingerPaths[id]);
    mFingerPaths[id].computeBounds(mPathBounds, true);
    invalidate((int) mPathBounds.left, (int) mPathBounds.top,
      (int) mPathBounds.right, (int) mPathBounds.bottom);
    mFingerPaths[id] = null;
  }
  for (int i = 0; i < cappedPointerCount; i++) {
    if (mFingerPaths[i] != null) {
      int index = event.findPointerIndex(i);
      mFingerPaths[i].lineTo(event.getX(index), event.getY(index));
      mFingerPaths[i].computeBounds(mPathBounds, true);
      invalidate((int) mPathBounds.left, (int) mPathBounds.top,
        (int) mPathBounds.right, (int) mPathBounds.bottom);
    }
  }
  //Create and set count of finger : End
```

However, there is limitation of the software. In terms of computing the average reading speed easily, we have to select one finger per hand at most, which means that we have to select the reader’s preferred reading finger.

V. A PILOT STUDY



Figure 5. Female participant using 2 hands in reading.

A pilot study was conducted for testing the application. Two participants, one male aged 16 and one female aged 62, were recruited from the School for the Blind, Bangkok, Thailand. They were informed that they could leave or stop the study anytime they wanted. The story used in reading is called “Sim Bai” converted from the Thai writing system comprising 40 lines of text to Thai braille consisting of 114 lines. The maximum cell numbers per line was 28. At the end of the story, the participants were asked to answer a few questions to check their comprehension. The study found that both readers used two hands in reading and they read aloud even though they were told they could read silently, they said this helped them to spell out words.

The female participant in figure 5 used a butterfly reading style; left and right hands conjoint at the beginning of the line then split near the middle, the right hand moves until the end of that line while the left hand moves to the beginning of a new line as shown in figure 6.

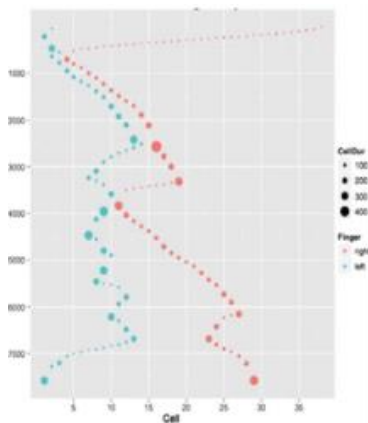


Figure 6. An example of a butterfly reading style from Aranyanak and Reilly [6].

The participant took 12.5 minutes to finish the reading. The male braille readers also used both hands in reading, but he preferred reading with both index fingers side-by-side along the line from the beginning to the end of the line. He took 10 minutes to finish the text. After the reading study, there was an interview session for about 10 minutes. Both readers said that they preferred reading on the transparent paper to normal paper because dots were not too sharp and not too rough, so it was more comfortable for them to read for longer. In addition, they were taught to read braille using both hands together in reading, however, they said sometimes they used only one finger in reading if the other gets tired and they had both developed a reading style suited to their specific needs and experience.

VI. CONCLUSION AND FUTURE WORK

This paper presents a new finger-tracking system developed on Android using Java. Finger movements can be used to investigate moment-to-moment processing of braille text. The device used is mobile and convenient to collect

data using only a tablet with the installed program. The application can show reading behavior of braille readers in real time. Cognitive processes can be inferred from the reading patterns found on the application screen. The software provides various reading time measures. Data is logged to a cloud-based database. The pilot study revealed that the finger tracking system for two fingers does not always provide an accurate trace, since the signals from both hands were sometimes confused. Nonetheless, data from single handed tracking proved robust and accurate.

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